

```
In [1]: if True: # indentation is always 4 spaces
        print('Data Science')
```

Data Science

```
In [2]: if True: # indentation is always 4 spaces
        print('Data Science')
```

```
Cell In[2], line 2
    print('Data Science')
    ^
```

IndentationError: expected an indented block after 'if' statement on line 1

```
In [3]: if True: # indentation is always 4 spaces
        print('Data Science')
```

Data Science

```
In [4]: if False:
        print('Data Science')
        print('bye for now')
```

bye for now

```
In [7]: if True:
        print('Data Science')
        print('bye for now')
```

Data Science

bye for now

```
In [5]: if True:
        print('Data Science')
        else:
        print('bye for now')
```

Data Science

check the number is even number or odd number

```
In [6]: x = 4
        r = x % 2

        if r == 0:
            print('Even number')
```

Even number

```
In [7]: x = 5
        r = x % 2

        if r == 0:
            print('Even number')
```

```
In [11]: x = 5
         r = x % 2
```

```
if r == 0:  
    print('Even number')  
  
print('odd number')
```

odd number

```
In [8]: x = 8  
        r = x % 2  
  
        if r == 0:  
            print('Even number')  
  
        print('odd number')
```

Even number

odd number

```
In [9]: x = 8  
        r = x % 2  
  
        if r == 0:  
            print('Even number')  
  
        if r == 1:  
            print('odd number')
```

Even number

```
In [14]: x = 5  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
  
         if r == 1:  
             print('odd number')
```

odd number

```
In [16]: x = 37  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
  
         if r == 1:  
             print('odd number')
```

odd number

```
In [17]: x = 37  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
  
         if r == 1:  
             print('odd number')  
  
         if r == 2:  
             print('evenn')
```

```
if r == 3:  
    print('odd')
```

odd number

```
In [19]: x = 10  
r = x % 2  
  
if r == 0:  
    print('Even number')  
else:  
    print('odd number')
```

Even number

please check number is even number or odd
numebr

please chenk number is greater or smaller

```
In [20]: x = 3  
r = x % 2  
  
if r == 0:  
    print('Even number')  
  
    if x>5:  
        print('greater number')  
  
else:  
    print('Odd Number')
```

Odd Number

```
In [21]: x = 4  
r = x % 2  
  
if r == 0:  
    print('Even number')  
  
    if x>5:  
        print('greater number')  
  
else:  
    print('Odd Number')
```

Even number

```
In [22]: x = 4  
r = x % 2  
  
if r == 0:  
    print('Even number')  
  
    if x>5:  
        print('greater number')  
    else:  
        print('smaller number')
```

```
else:  
    print('Odd Number')
```

Even number
smaller number

```
In [10]: x = 10  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
  
             if x>5:  
                 print('greater number')  
             else:  
                 print('smaller number')  
  
         else:  
             print('Odd Number')
```

Even number
greater number

```
In [11]: x = 4  
  
         if x == 1:  
             print('one')  
         if x == 2:  
             print('Two')  
         if x == 3:  
             print('Three')  
         if x == 4:  
             print('four')
```

four

```
In [28]: x = 4  
  
         if x == 1:  
             print('one')  
         elif x == 2:  
             print('Two')  
         elif x == 3:  
             print('Three')  
         elif x == 4:  
             print('four')
```

four

```
In [12]: x = 7  
  
         if x == 1:  
             print('one')  
         elif x == 2:  
             print('Two')  
         elif x == 3:  
             print('Three')  
         elif x == 4:  
             print('four')
```

```
In [13]: x = 40

if x == 1:
    print('one')

elif x == 2:
    print('Two')
elif x == 3:
    print('Three')
elif x == 4:
    print('four')

else:
    print('number not found')
```

number not found

loops (16th

```
In [14]: i = 1          # initializing

while i<=5:          # condition
    print('data science')
    i = i + 1        # increment
```

data science
data science
data science
data science
data science

```
In [15]: i = 1          # initializing

while i<=5:          # condition
    print('data science',':', i)
    i = i + 1        # increment
```

data science : 1
data science : 2
data science : 3
data science : 4
data science : 5

```
In [16]: i = 1          # initializing

while i>=1:          # condition
    print('data science',':', i)
    i = i - 1        # increment
```

data science : 1

```
In [9]: # nested while loop

i = 1

while i<=5:
    print('data science') # when we mention end then new line will not create
    j = 1
    while j<=4:
```

```

    print('technology')
    j = j + 1

    i = i + 1
    print()

```

the output which we got is very lengthy but how to make them one line Lets

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

In [17]: *# nested while loop*

```

i = 1

while i<=5:
    print('data science', end = ' *^ ') # when we mention end then new line will
    j = 1
    while j<=4:
        print('technology', end = ' *^ ')
        j = j + 1

    i = i + 1
    print()

```

the output which we got is very lengthy but how to make them one line Lets

```

data science *^ technology *^ technology *^ technology *^ technology *^
data science *^ technology *^ technology *^ technology *^ technology *^
data science *^ technology *^ technology *^ technology *^ technology *^
data science *^ technology *^ technology *^ technology *^ technology *^
data science *^ technology *^ technology *^ technology *^ technology *^

```

In [18]: *# Lets use while loop usig some numbers*

```
i = 1

while i <= 2 :
    j = 0
    while j <= 2 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

0 1 2

0 2 4

In [15]: *# Lets use while loop usig some numbers*

```
i = 1
while i <= 4 :
    j = 0
    while j <= 3 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

0 1 2 3

0 2 4 6

0 3 6 9

0 4 8 12

for loop

In [19]: name = 'nit'

```
for i in name:
    print(i)
```

n

i

t

In [20]: name = [1,2,3]

```
for i in name:
    print(i)
```

1

2

3

In [21]: name = (1,2,3)

```
for i in name:
    print(i)
```

1

2

3

```
In [22]: range(5)
```

```
Out[22]: range(0, 5)
```

```
In [23]: for i in range(5):  
         print(i)
```

```
0  
1  
2  
3  
4
```

```
In [24]: for i in range(1,10,3):  
         print(i)
```

```
1  
4  
7
```

```
In [23]: # print the number which divisibe by 5
```

```
for i in range(1,51):  
  
    if i%5==0 :  
        print(i)
```

```
5  
10  
15  
20  
25  
30  
35  
40  
45  
50
```

- break
- continue
- pass

```
In [28]: import keyword  
keyword.kwlist
```

```
Out[28]: ['False',
          'None',
          'True',
          'and',
          'as',
          'assert',
          'async',
          'await',
          'break',
          'class',
          'continue',
          'def',
          'del',
          'elif',
          'else',
          'except',
          'finally',
          'for',
          'from',
          'global',
          'if',
          'import',
          'in',
          'is',
          'lambda',
          'nonlocal',
          'not',
          'or',
          'pass',
          'raise',
          'return',
          'try',
          'while',
          'with',
          'yield']
```

```
In [24]: for i in range(1,11):
          print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

```
In [25]: for i in range(1,11):
          if i == 3:
              break
          print(i)
```

```
1
2
```

```
In [32]: for i in range(1,11):
          if i == 3:
```

```

    continue
    print(i)

```

```

1
2
4
5
6
7
8
9
10

```

In [36]: `for i in range(1,11):`

```

Cell In[36], line 1
    for i in range(1,11):
        ^
_IncompleteInputError: incomplete input

```

In [34]: `for i in range(1,11):`
`pass`

In [37]: *# write the code user ask choclet from vendor machine write the basic code*

```

x = int(input('How many choclets you want:?'))

i = 1
while i<=x:
    print('choclet')
    i += 1

```

```

choclet
choclet
choclet
choclet
choclet

```

In [38]: *ava = 5 # the machine has only 5 choclet*

```

x = int(input('How many choclets you want:?'))

i = 1
while i<=x:
    print('choclet')
    i += 1

```

```

choclet
choclet
choclet
choclet
choclet
choclet
choclet
choclet
choclet
choclet

```

In [40]: *available_choclet = 5 # the machine has only 10 candis*

```

x = int(input('How many choclets user want:?'))

```

```

i = 1
while i<=x:

    if i>available_choclet: # we stop the execution but which code execution not
        break # break is statement | means jump out of the loop
    print('choclet')
    i += 1

print('out of stock & bye for now')

```

choclet
choclet
choclet
choclet
choclet
out of stock & bye for now

```

In [26]: available_choclet = 5 # the machine has only 10 candies

x = int(input('How many chocolates you want:'))

i = 1
while i<=x:

    if i>available_choclet: # we stop the execution but which code execution not
        print('out of stock')
        break # break is statement | means jump out of the loop
    print('choclet')
    i += 1

print('bye for now')

```

choclet
choclet
choclet
choclet
bye for now

**you need to print the number from 1 to 50 but
dont print the number which is divisible by 3 or 5**

```

In [27]: for i in range(1,51):

    if i%3 == 0:
        #continue

    print(i)
print('end')

```

3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
end

```
In [45]: for i in range(1,51):  
        if i%3 == 0:  
            continue  
        print(i)  
    print('end')
```

1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
end

```
In [30]: for i in range(1,51):  
        if i%3 == 0 or i%5 == 0:  
            continue  
        print(i)  
        #print('end')  
        # it will skip all the value which is divisible by 3 or 5
```

1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49

```
In [33]: a = input('enter the value').split()  
        print(a)
```

['thank', 'you', 'sir']

```
In [37]: a = input('enter the value').split()  
        num = [int(i) for i in a]  
        print(num)
```

[10, 20, 30]

In []:

```
In [55]: a = input('enter the value')  
        vol = 'aeiou'  
        checking = sum([1 for i in a if i.lower() in vol])  
        print(f'Input word is {a} and the no of vowels we used {checking}')
```

Input word is aechgfo and the no of vowels we used 3

```
In [62]: num = int(input("Enter a number: "))  
        print("Square:", num ** 2)
```

Square: 4

```
In [65]: a = input('enter the value')
        b = a.split(',')
        print(b)
```

```
['asd', 'dfdfd', 'dfdfs']
```

```
In [67]: user_input = input("Enter True or False: ").lower() == "true"
```

```
In [71]: num = int(input("Enter a number: "))
        factorial = 1
        for i in range(1, num + 1):
            factorial *= i
        print("Factorial:", factorial)
```

```
Factorial: 120
```

```
In [78]: num = input("Enter a number: ").strip()
        print(num)
```

```
In [ ]:
```

```
In [ ]:
```